

Read each question carefully and completely. Be sure your answer addresses the question being asked. Use complete sentences.

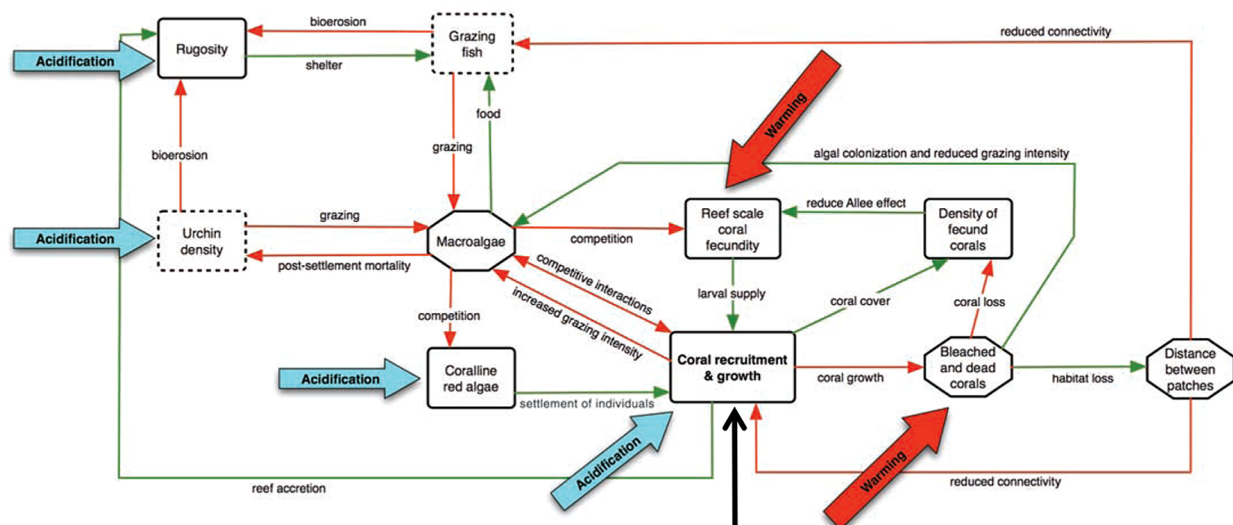
- (2 pts) 1. Name the type of bubbles that snapping shrimp and mantis shrimp use to stun their prey.
- (4 pts) 2. Name two types of reproduction (giving birth) in sharks.
- (4 pts) 3. Name two ways that sponges can reproduce asexually.
- (5 pts) 4. Explain one reason a manager must understand how genetic variation is distributed across the entire range of a species that is being considered for conservation management or reintroduction into a particular area.

- (10 pts) 5. Write the equilibrium equation for the carbonate buffering system in marine ecosystems. Use this equation to explain how the increase in atmospheric CO_2 increases ocean acidity. Circle the hydrogen ion that contributes to most increase in acidity.

- (5 pts) 6. Write the equilibrium equation for CaCO_3 chemistry.

- (10 pts) 7. Explain how the increased acidity (from the equilibrium equation in question 5) interacts with CO_3^{2-} (from the equilibrium equation in question 6) to reduce CaCO_3 availability to marine animals. Include the proper chemical equation as part of your answer

- (5 pts) 8. What is the minimum value of Ω necessary for carbonate skeleton formation? What is the optimum value?



- (20 pts) 9. Use the figure above (also on screen) to clearly and thoroughly explain how increased ocean acidity will alter connectivity and fecundity across a large coral reef. You must include coral recruitment and growth (indicated by the black arrow), distance between patches and coral fecundity. Be sure to include both positive and negative relation among these factors. Do not include unnecessary factors and relationships.

(10 pts) 10. Briefly explain each of the three factors that must be considered to identify a potential biodiversity hotspot for conservation efforts.

11. Select whether you want **2 points** or **5 points** added to your score for this exam. Before you decide, there is a small catch: if more than 10% of the class selects 5 points, then no one gets the extra points. Only I will see your choice.

A. 2 points. B. 5 points.